



# Bubbles

AI Powered Communication Assistant

Group Members

**Muhammad Ahmed & Attique Rehman Alias Adeel**

Supervisor:

**Dr. Jawad Shafi**

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# 1.0 Introduction & Importance

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# Communication Skills Importance

Good communication serves as the foundational cornerstone for professional success, academic achievement, and everyday social interactions. In our fast-paced world, the ability to articulate concepts clearly, understand subtleties in tone, and retain vital information from spoken conversations is essential for achieving your goals



# Decline in Communication skills

**01** Missed Key Details

**02** Agenda Drift

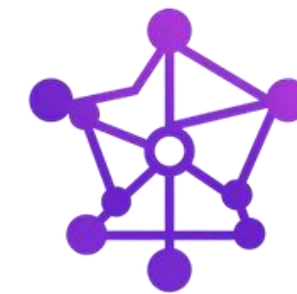
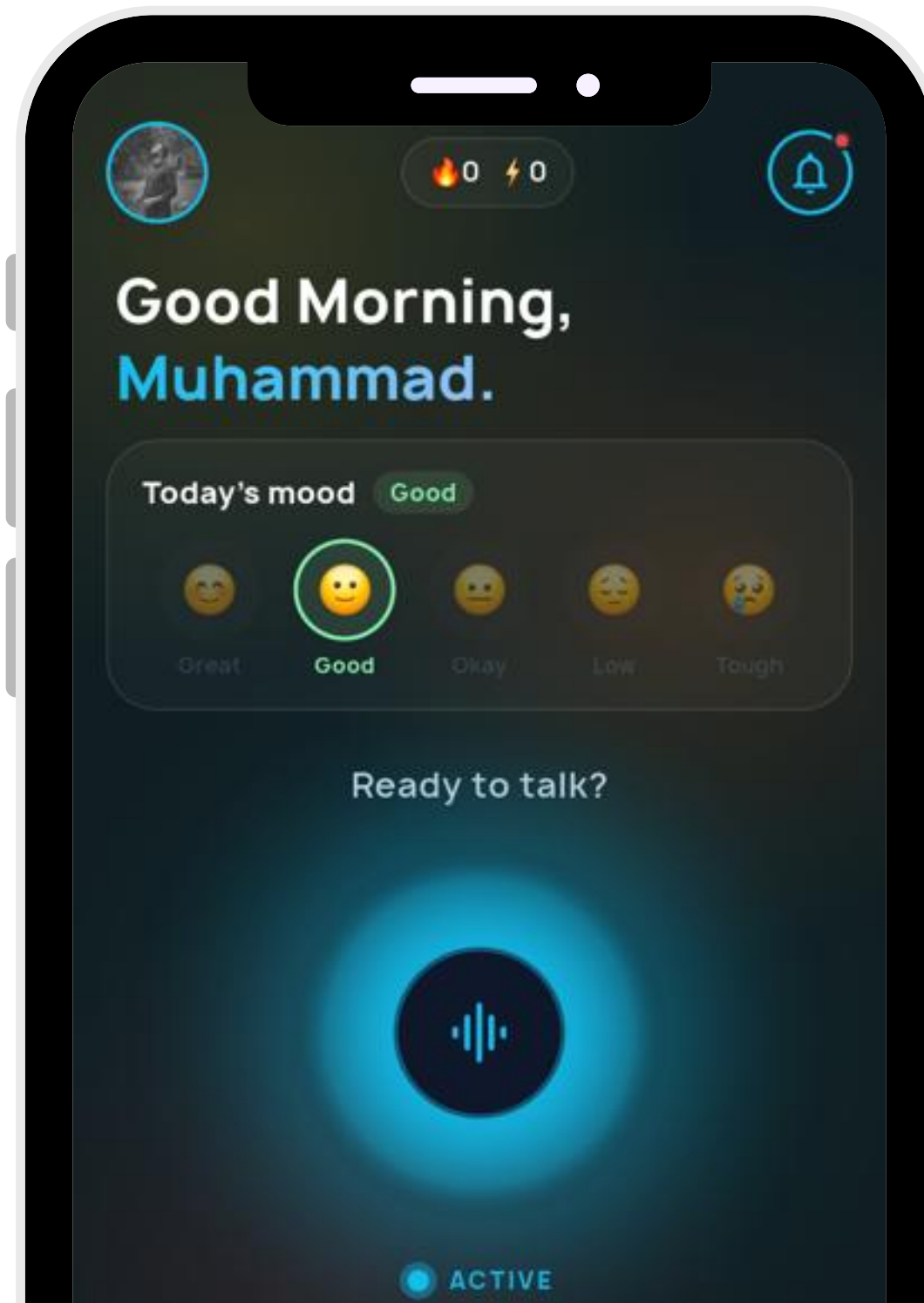
**03** Post-Conversation Regret

**04** Lack of Real-Time Feedback

**05** Misaligned Key Concepts

**06** Inappropriate Tone Usage

# Proposed Solution



# 01

## Live Coaching

Provides instant, in-session feedback on tone, clarity, and phrasing to help users identify and correct communication errors while they are actively speaking



# 02

## Contextual Q&A:

Employs a Knowledge Graph and vector memory to allow users to accurately search and ask questions about people, topics, and details from past conversations

# Core Objectives

# 03

## Gamified Growth

Motivates users to consistently improve their communication habits through an engaging progress dashboard featuring XP, levels, streaks, and quests



# 04

## Roleplay Practice

Enables users to select and converse with specific entities (such as people) extracted from past sessions, allowing them to rehearse real-life scenarios and practice conversations



# 05

## Insights Dashboard

Displays events, key facts, and action items extracted automatically from conversations, allowing users to quickly review important details later and ask questions about each entity without having to read through full transcripts.

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# 2.0 Main Features

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## Live audio transcription

Instantly converts spoken words into text during conversations

## Active speaker diarization

Identifies and distinguishes between different speakers in real time

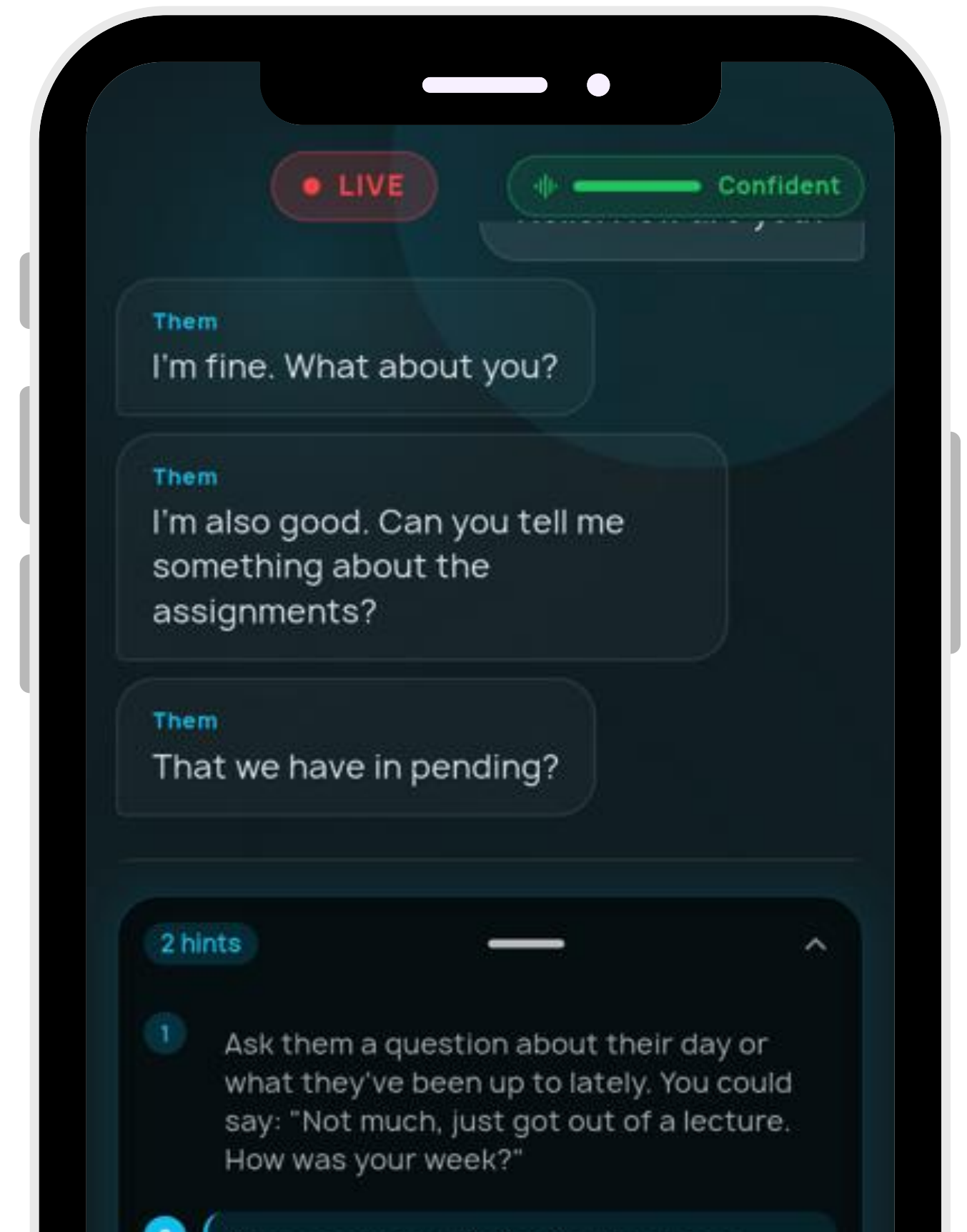
## Instant coaching suggestions

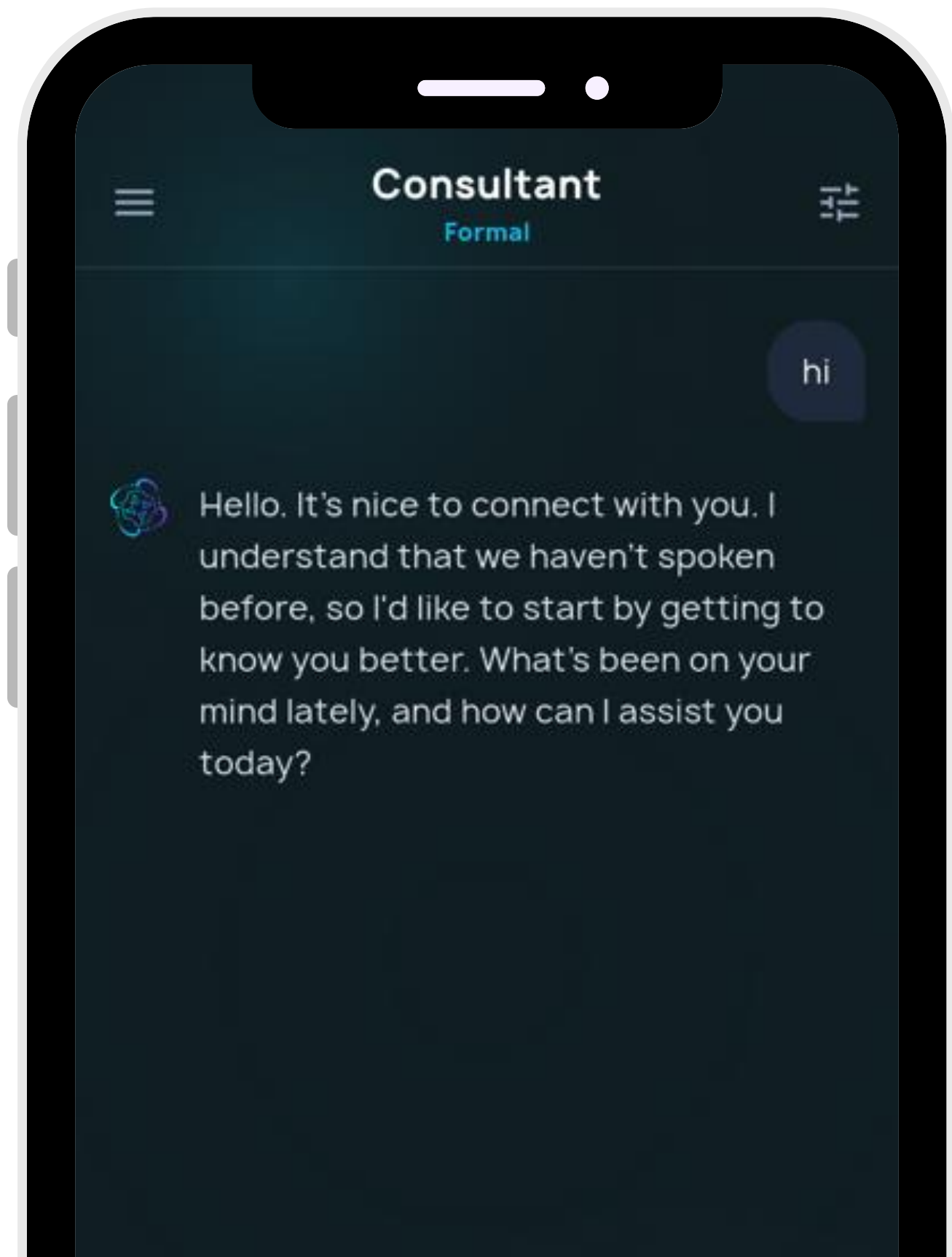
Provides immediate, actionable communication advice while you speak

## Analyzes conversation tone

Evaluates the emotional undertone of the ongoing dialogue

# Live Wingman





# Consultant Mode

## Context-aware Q&A

Gives grounded answers based on knowledge of your past sessions

## Queries past conversations

Searches through past dialogue data to retrieve relevant details

## Hybrid RAG retrieval

Merges vector similarity & graph search for accurate answers

## Text or voice input

Interact and ask questions by either typing or speaking

# Roleplay System

## Select graph entities

Choose a person or persona from knowledge graph to interact with

## Injects relational context

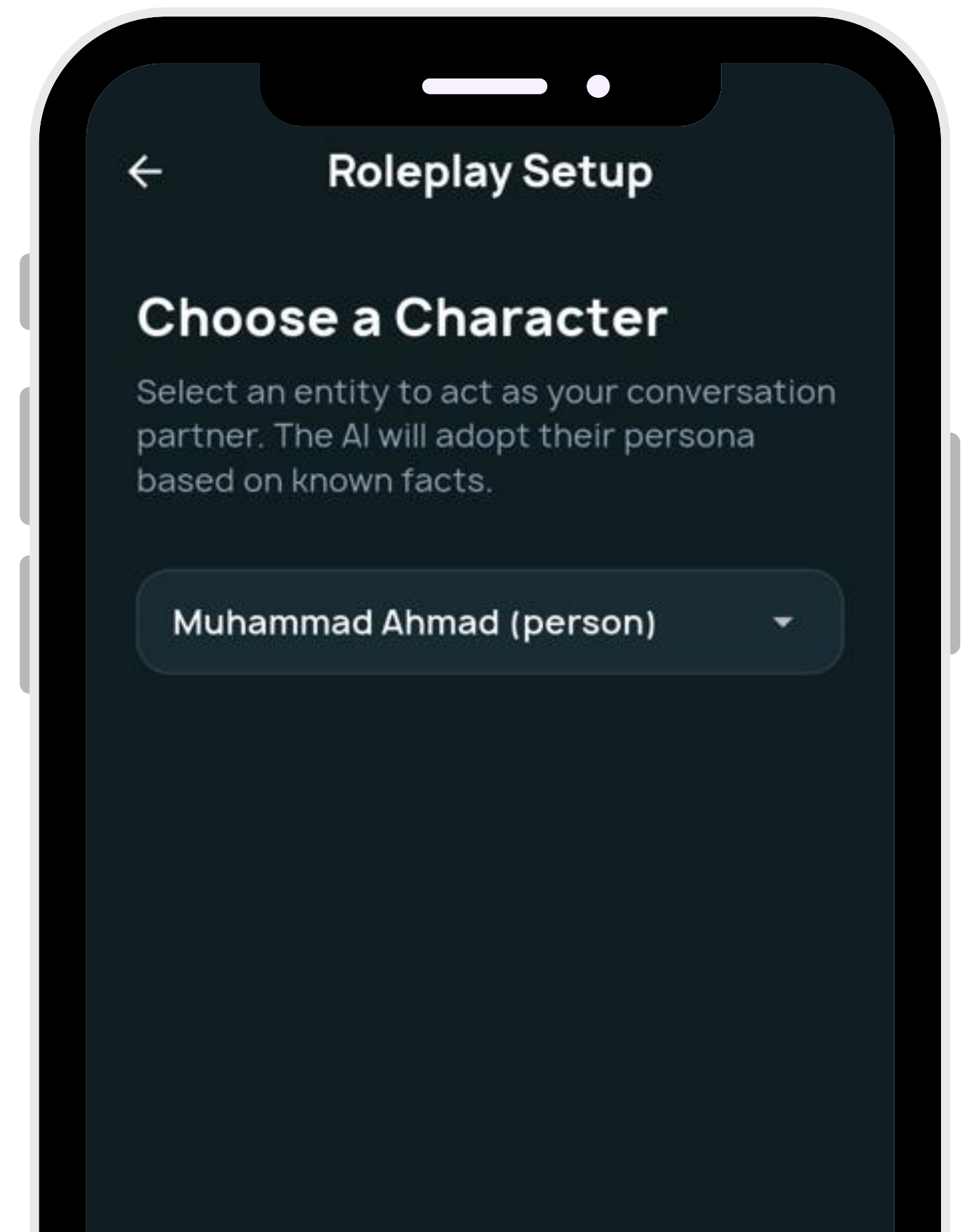
Fetch historical relationship data to make simulated conversation realistic

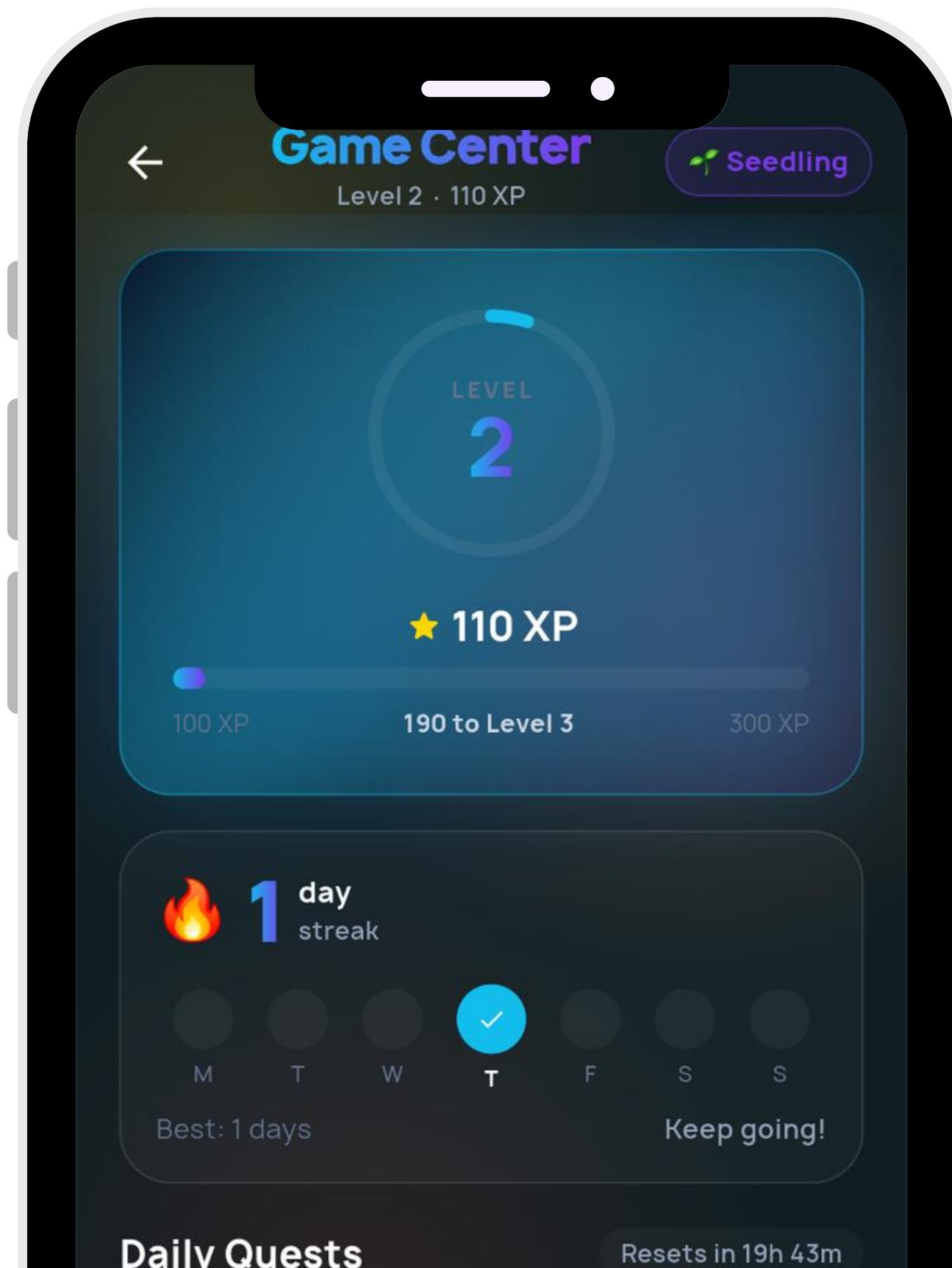
## Practice past personas

Rehearse conversations with individuals you've previously encountered

## Rehearse realistic scenarios

Simulates true-to-life interactions to help you prepare for actual meetings





## Earn XP and levels

Rewards experience points to level up based on application usage

## Complete daily customized quests

Offers personalized daily challenges to encourage users

## Track daily activity streaks

Monitors consecutive days of usage to help you build a habit

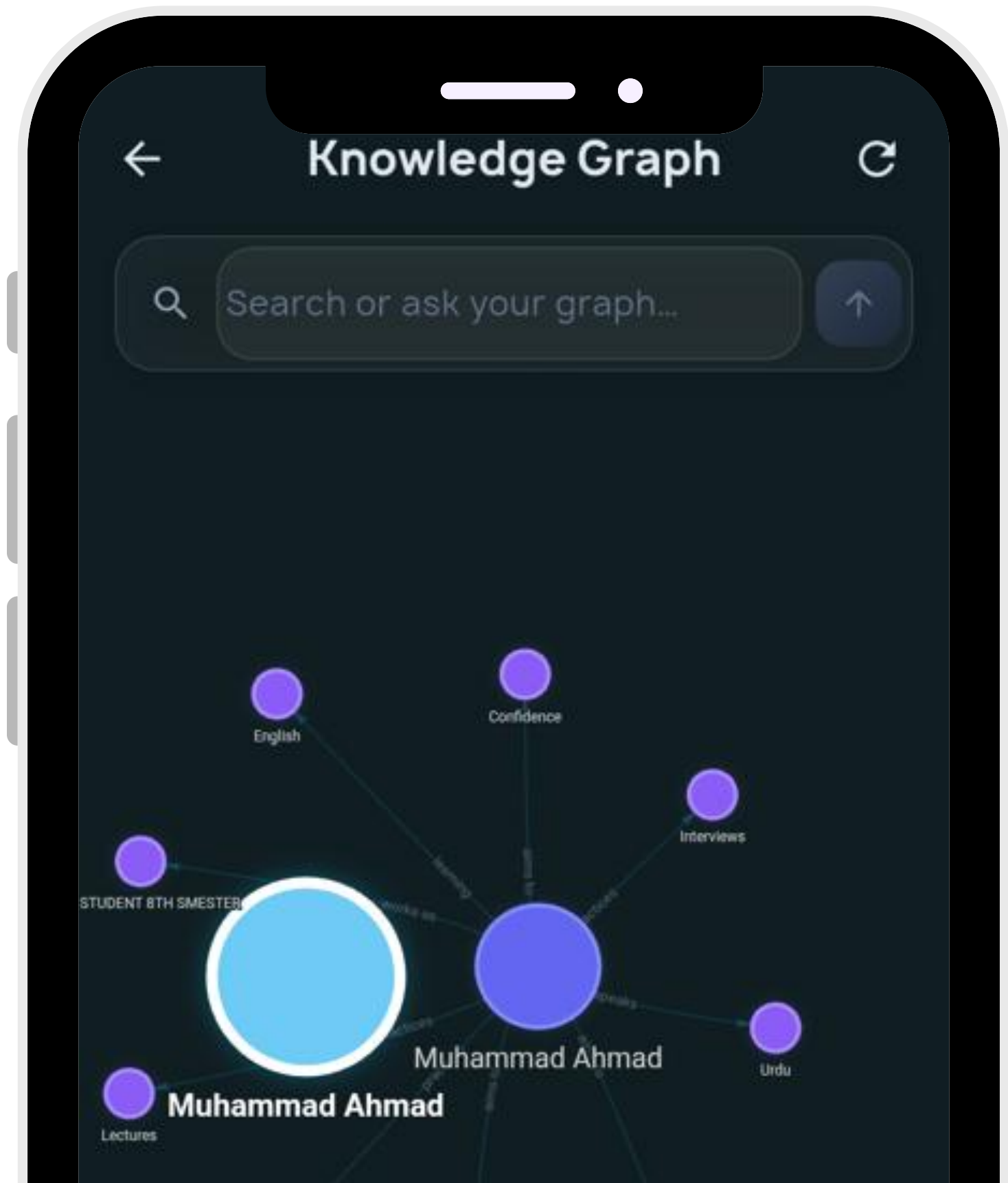
## Compete on user leaderboards

Ranks progress against other users to motivate friendly competition

# Gamification Engine

# KnowledgeGraph

## Explorer



# Knowledge Graph Explorer

## Interactive visual entity graph

Displays connections between people and topics in a dynamic visual web

## Tap for AI summaries

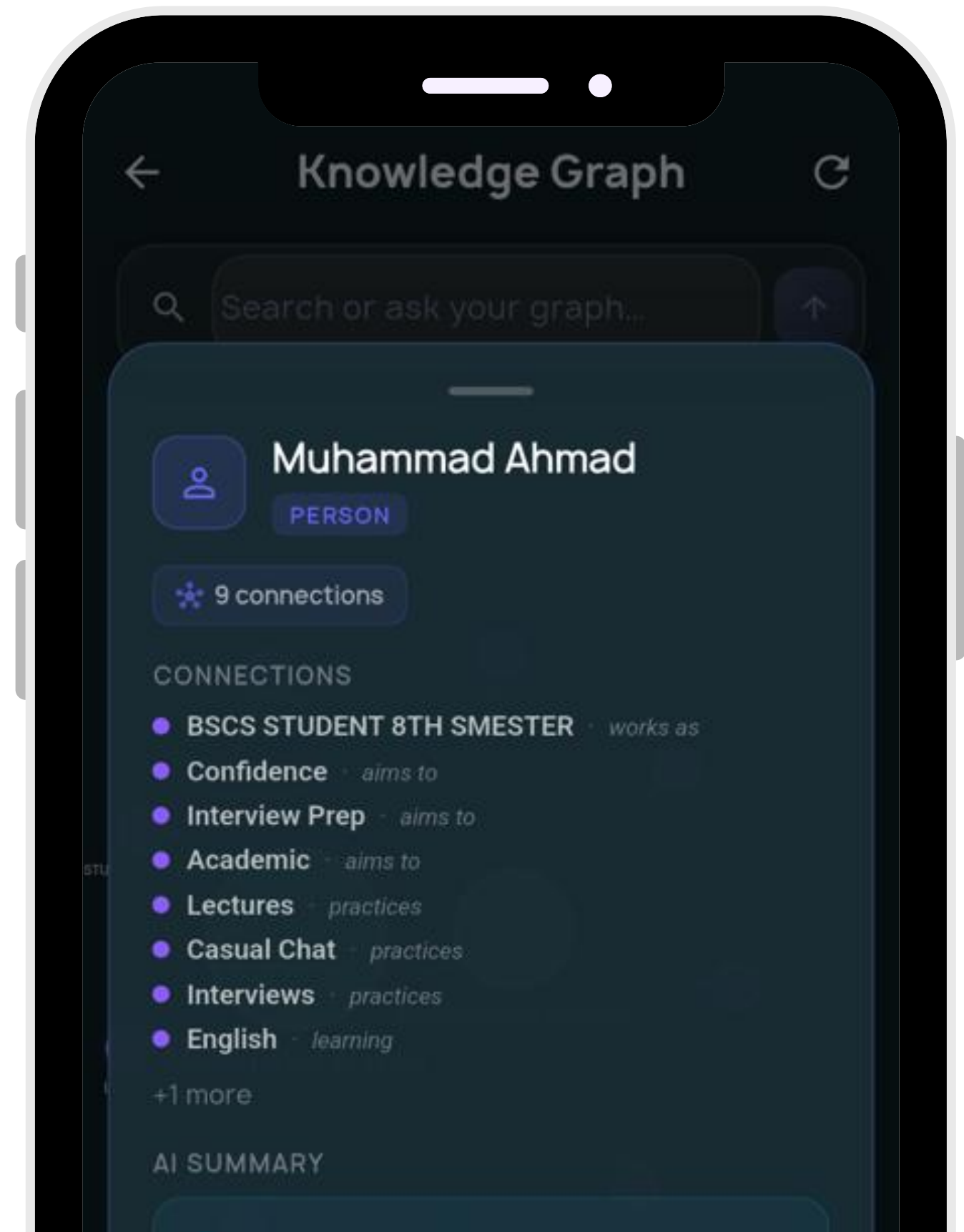
Generates instant AI overviews of an entity when you select its node

## Color-coded node types

Differentiates entity categories (like people or places) using distinct colors

## Explore connections

Allows intuitive navigation of relationships between different data points



# Voice Controls

## Hands-free app control

Enables navigation and use of application entirely via voice commands

## Custom wake word activation

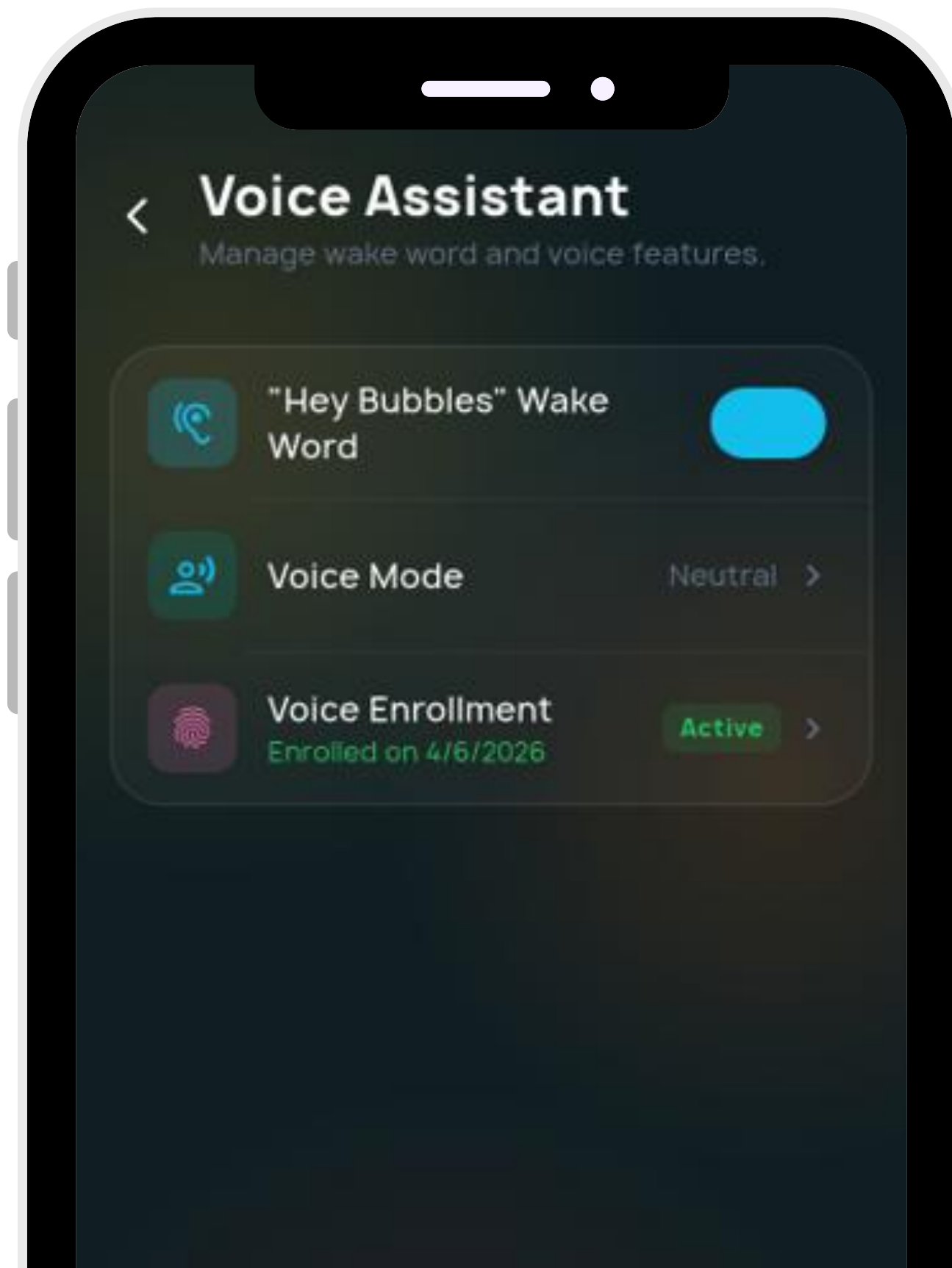
Uses the "Hey Bubbles" phrase to trigger the assistant seamlessly

## Background listening mode

Continuously listens for wake word without needing to tap screen

## LLM intent routing

Directs spoken commands to correct feature using LLM



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# 3.0 System Architecture

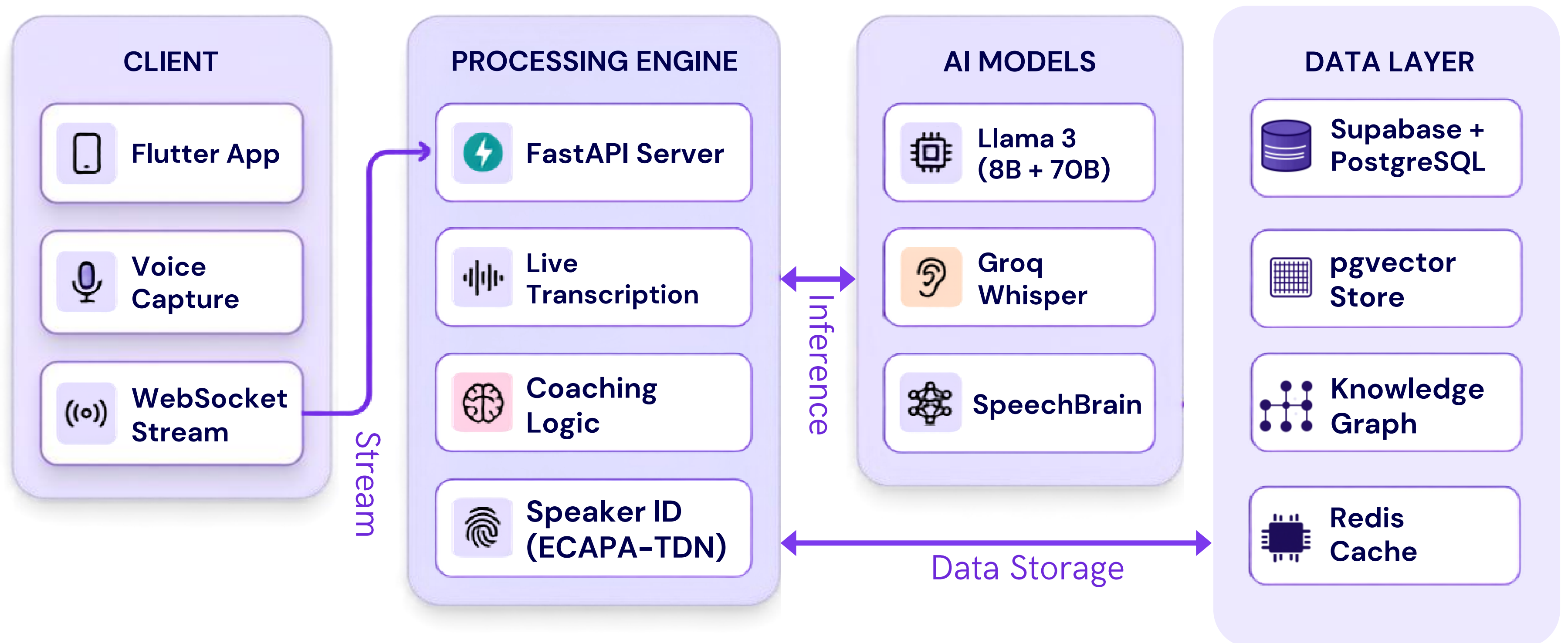
Group Members

**Muhammad Ahmed & Attique Rehman Alias Adeel**

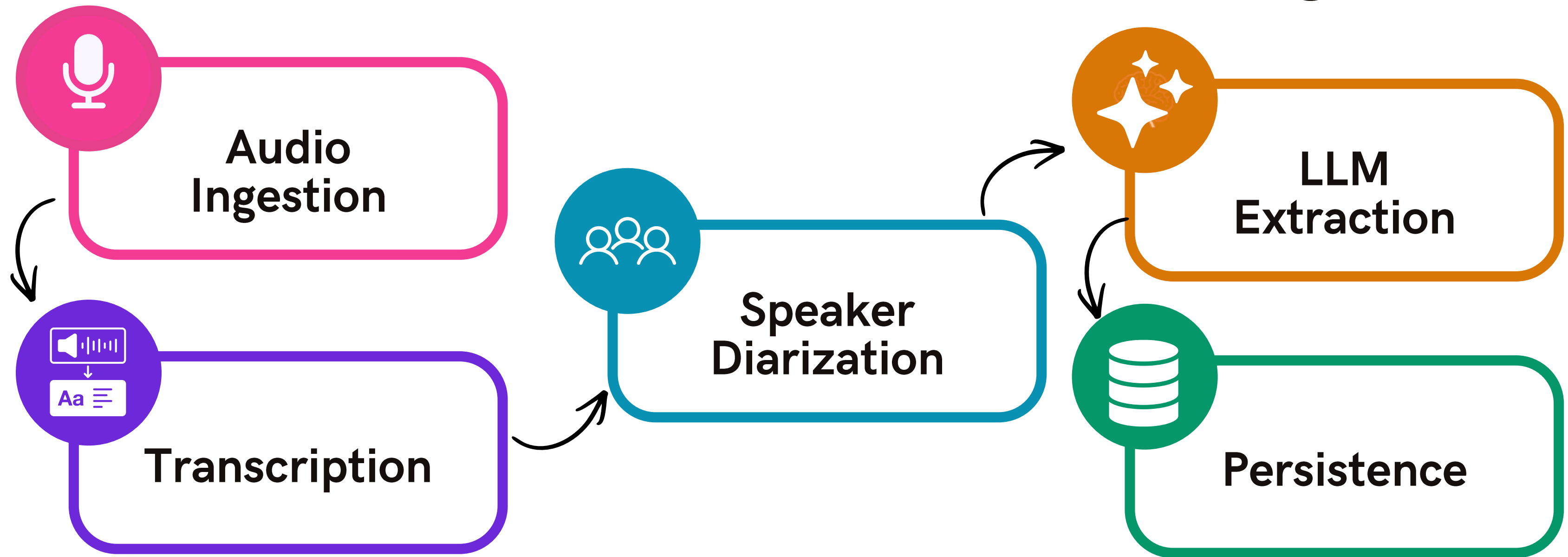
Supervisor:

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# Data Flow: From Audio to Insight



# ASR Model: Groq Whisper

Utilizes the whisper-large-v3-turbo

Uses Groq's high-speed Whisper model for accurate speech recognition

Low latency in text transcription

Converts speech to text with minimal delay to ensure real-time performance

Generates accurate timestamps

Precisely records the exact moment each word is spoken

Handles Voice Activity Detection

Automatically distinguishes between active speech and background silence



# Diarization Model

## SpeechBrain ECAPA

Utilizes speechbrain/spkrec-ecapa-voxceleb model (ECAPA-TDNN)

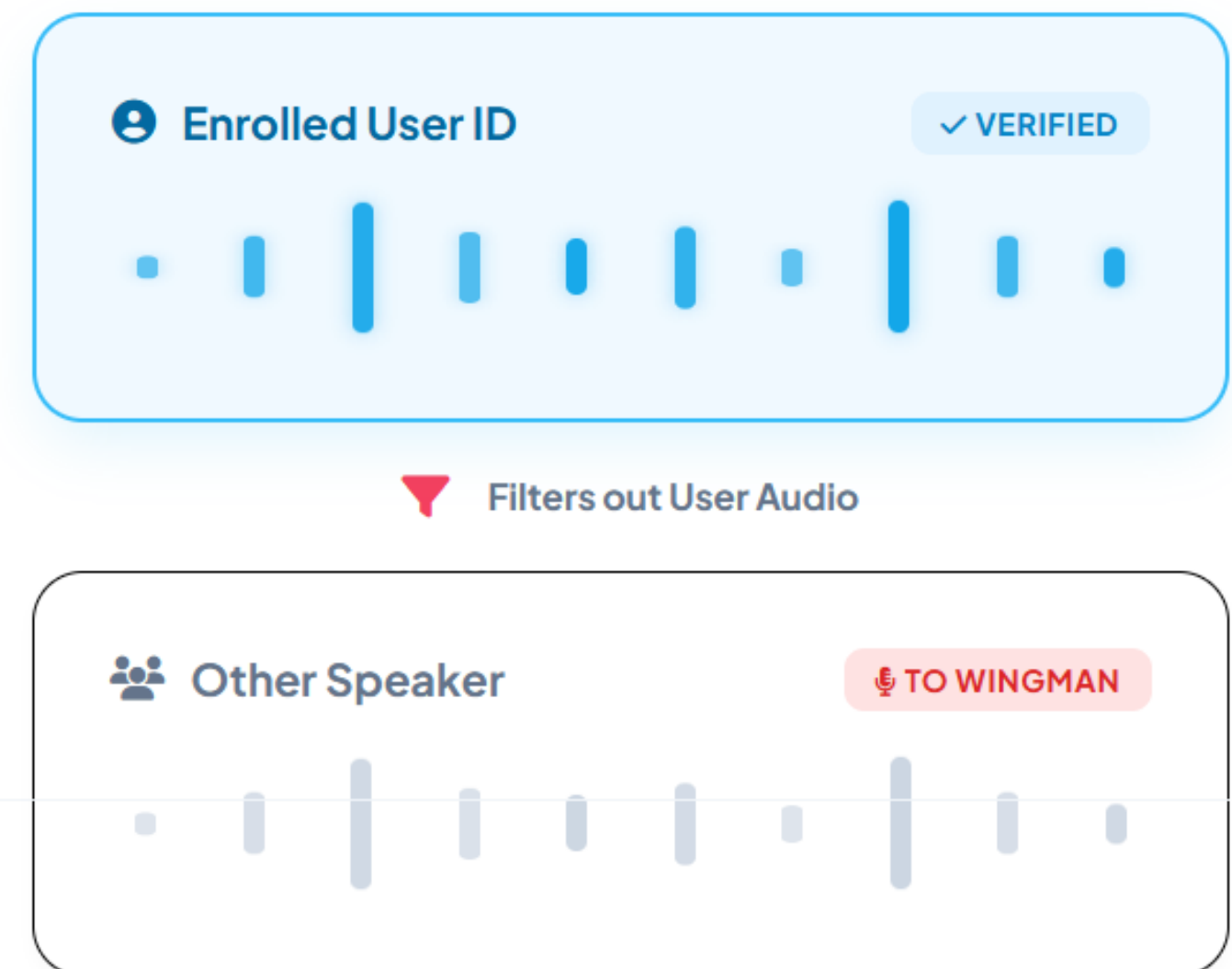
Employs ECAPA-TDNN neural network specialized for voice recognition

Tracking the enrolled user versus other

Differentiates your specific voice from other participants in the room

The model is lightweight and loads lazily

Designed for efficiency without consuming excessive resources



# Diarization Logic

## Vector Matching

Converts the user's enrolled voice sample to 192-dimensional embedding

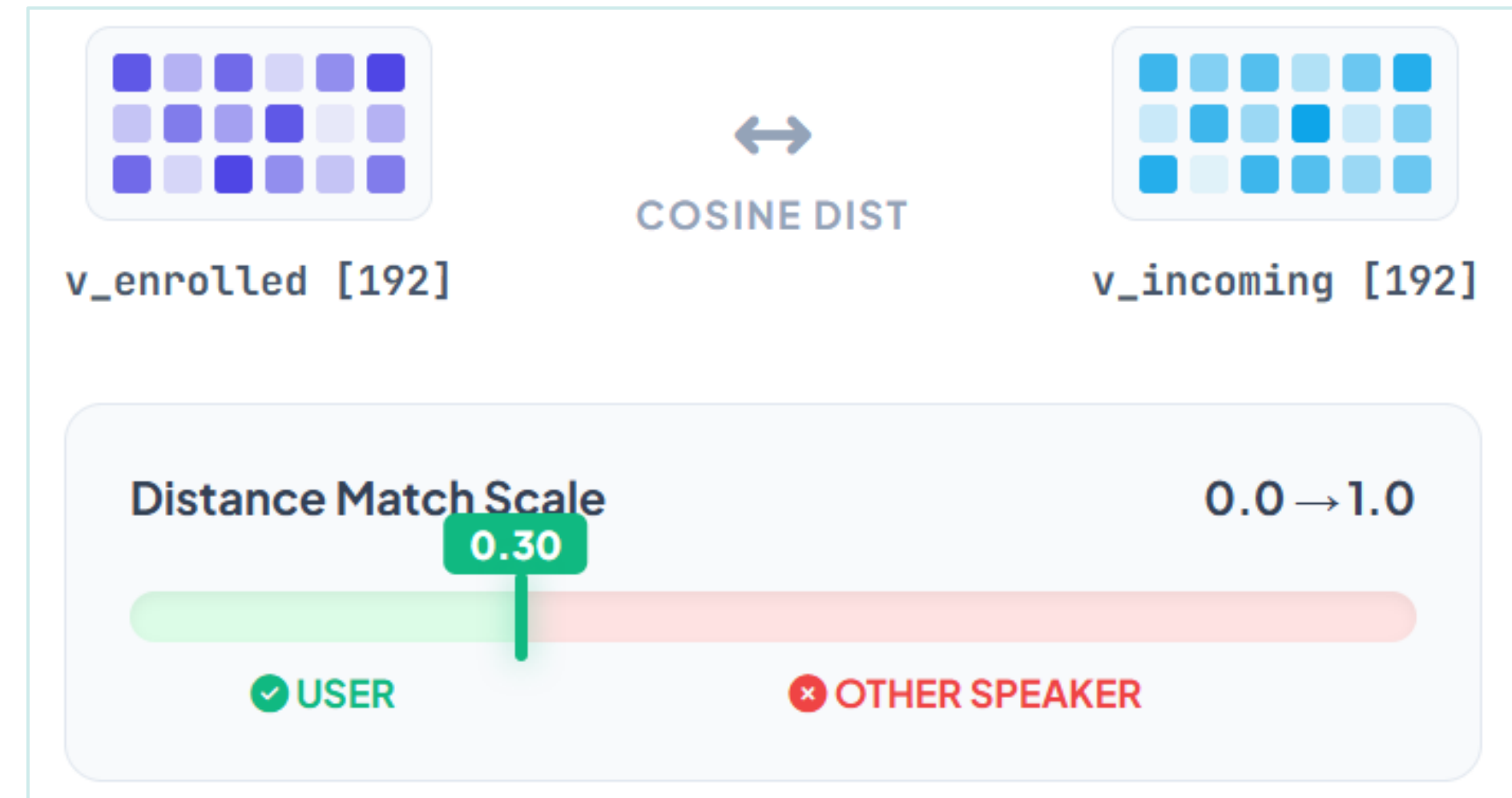
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Transforms your voiceprint into a dense mathematical vector

Audio chunks are matched against voice print using pgvector's cosine distance

---

Transforms your voiceprint into a dense mathematical vector



**Detection Precision**  
Verified against 30+ test sessions

**95%**

# Embedding Model: all-MiniLM

Uses the all-MiniLM-L6-v2

Employs efficient transformer model for creating text embeddings

Converts text into 384-dimensional vectors

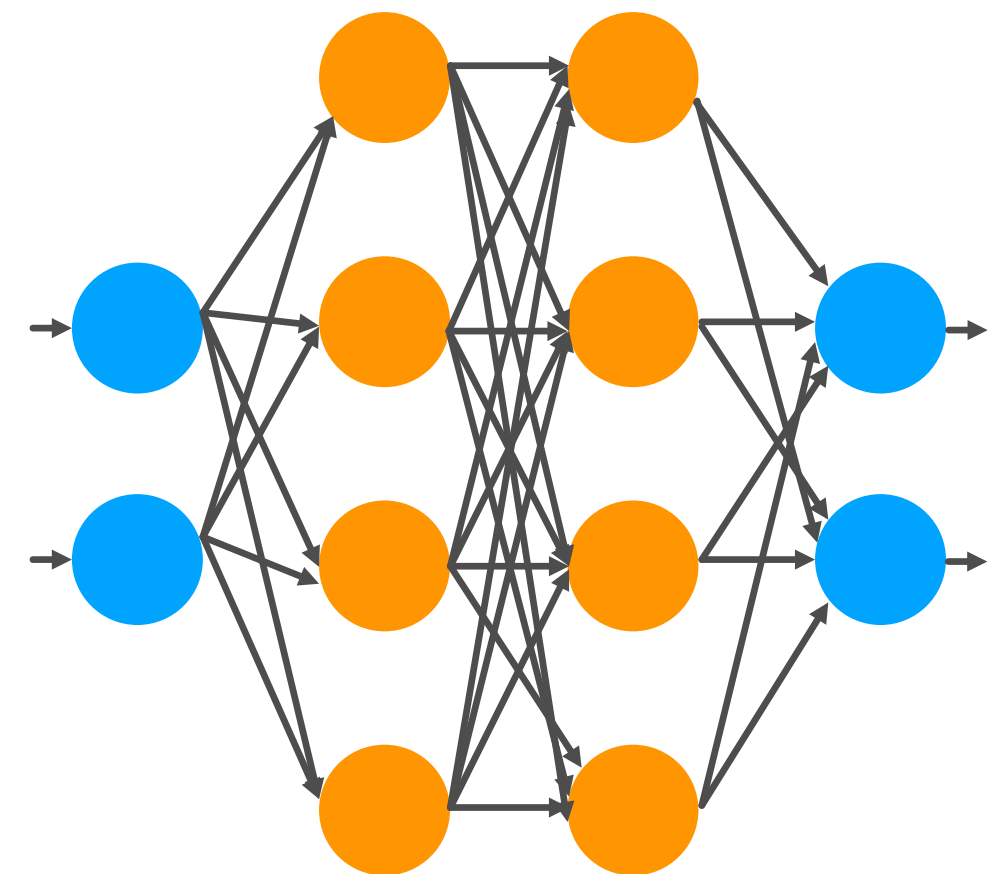
Transforms transcriptions into mathematical formats for semantic searchability

Operates locally on CPU resources

Runs directly on standard processors without requiring a dedicated GPU

Handles Voice Activity Detection

Assists in processing audio activity directly alongside the embeddings pipeline



# LLM Routing: Fast vs. Deep Inference

## Fast Tier Execution using Llama 3.1 8B

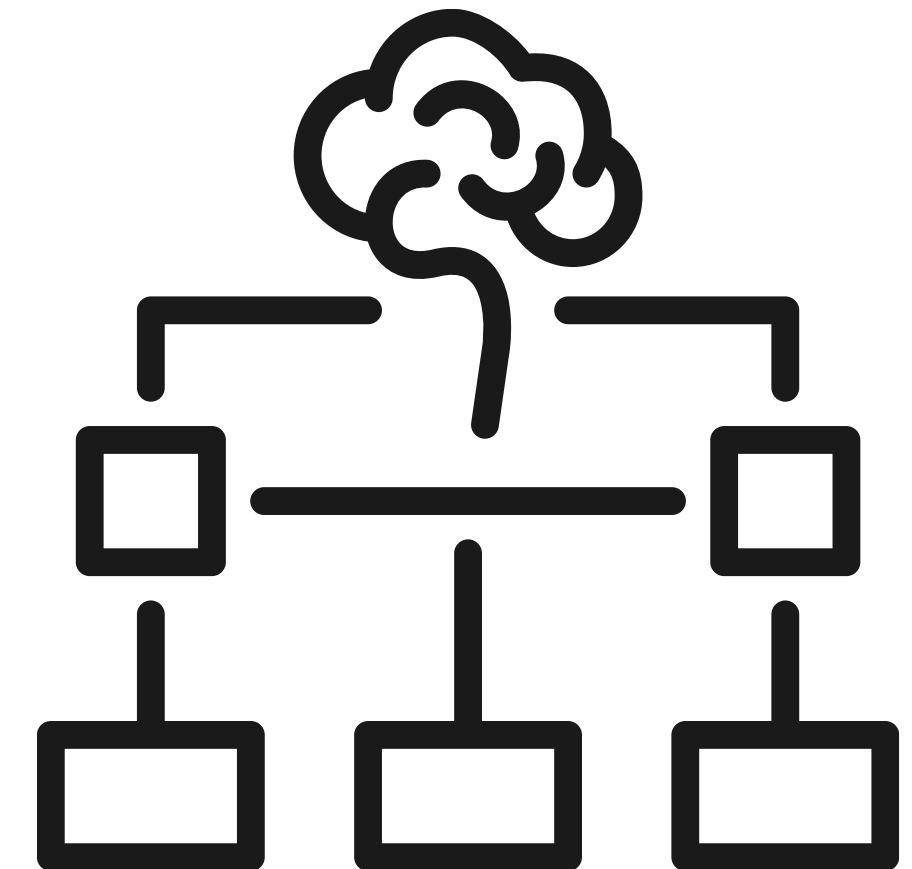
Uses a lightweight model for instant, low-latency Wingman advice

## Deep Tier Reasoning using Llama 3.3 70B

Uses a complex, heavy model for in-depth Consultant Q&A

## Reliability multi-provider Architecture

Implements fallback mechanisms across providers to ensure zero downtime



# Database: Architecture

## Session Logs

### Purpose

Turn-by-turn conversational record.

### Fields

Speaker identity, diarization labels, transcript confidence, inline sentiment scores.

## Vector Memory

### Purpose

Long-term semantic memory store.

### Fields

384-dim pgvector for cosine similarity search, importance weights, confidence tags.

## Graph Storage

### Purpose

Relational mapping and factual context.

### Fields

Canonical names (nodes), relationship strength (edges), specific factual details.

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# 4.0 Hybrid RAG

Group Members

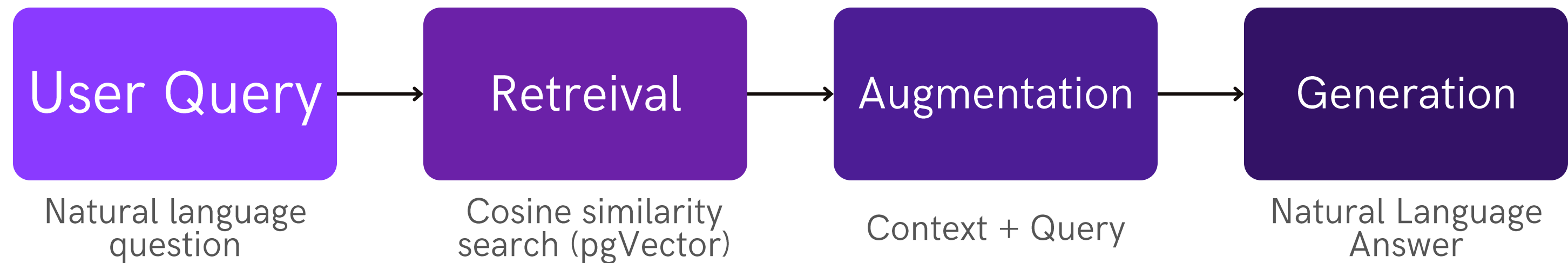
**Muhammad Ahmed & Attique Rehman Alias Adeel**

Supervisor:

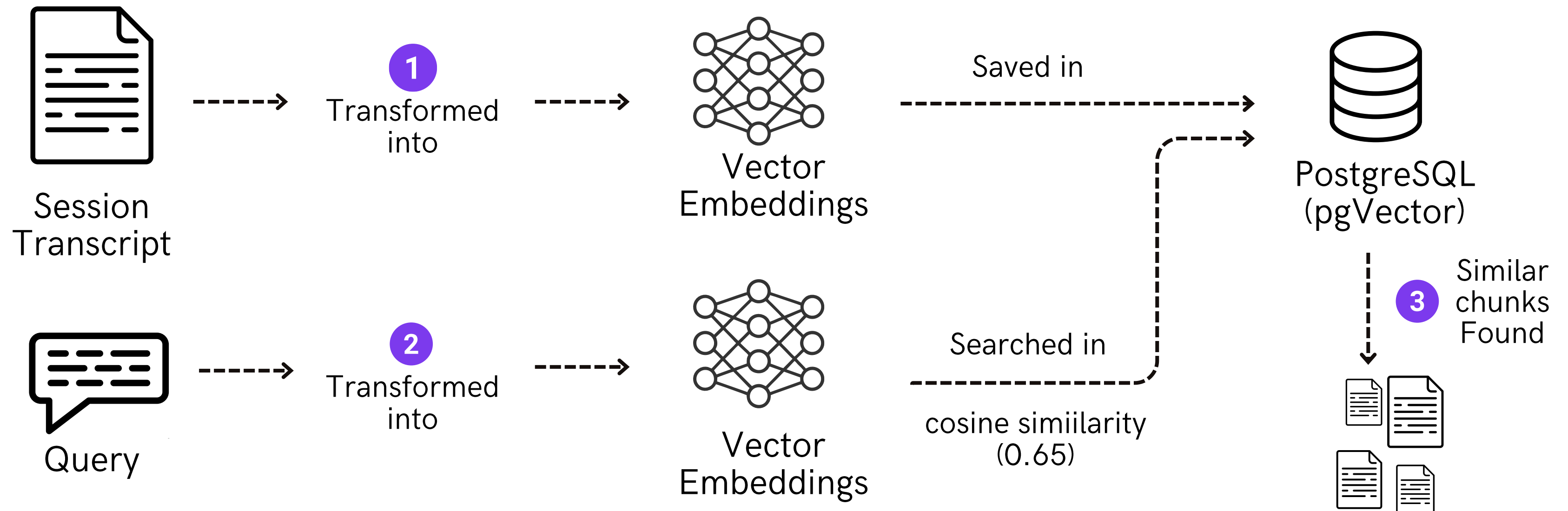
**Dr. Jawad Shafi**

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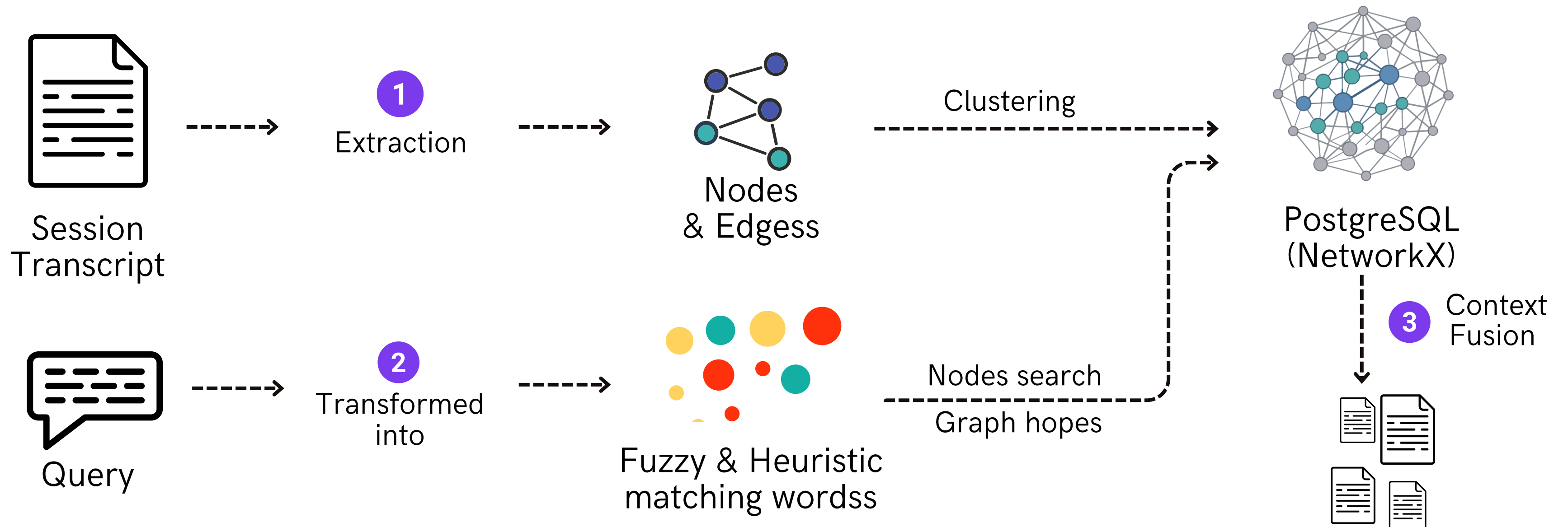
# What is: RAG



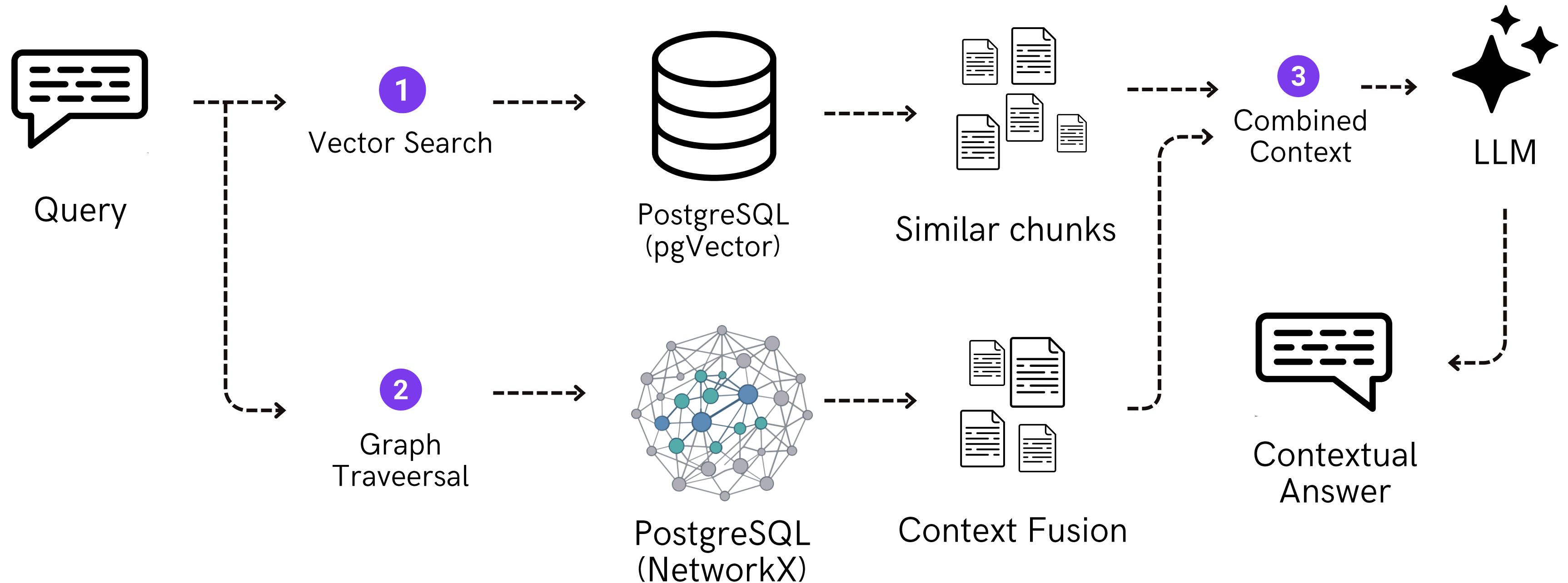
# Hybrid RAG: Vector Search



# Hybrid RAG: Graph Traversal



# Hybrid RAG: Fusing Data



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# 5.0 Implementation

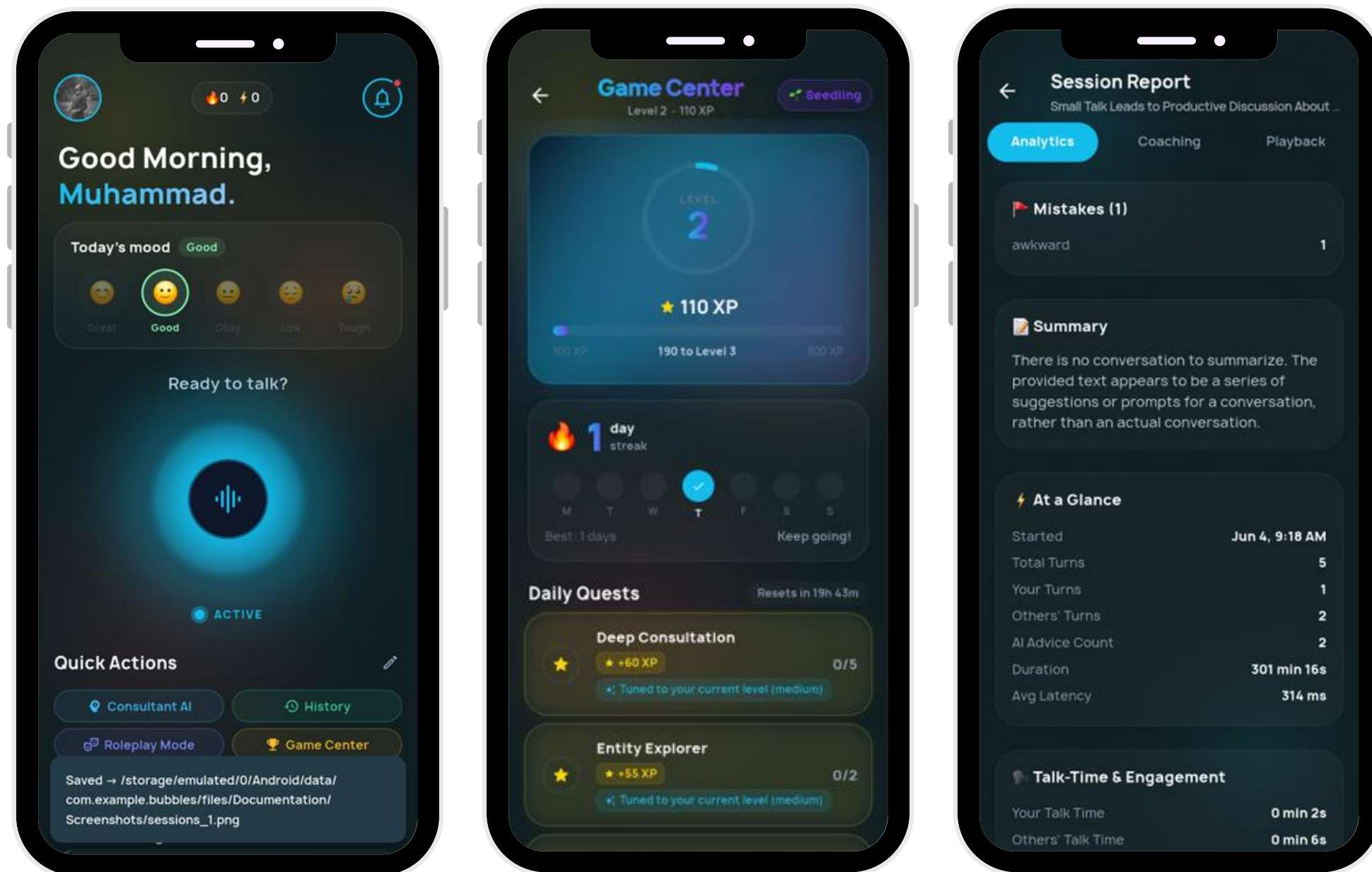
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## Clean UI using Flutter 3.10

Utilizes the Flutter framework for a smooth, cross-platform mobile experience



## Low-Latency Audio Pipeline

Maintains a direct, custom WebSocket connection to ensure real-time audio streaming and processing with minimal delay

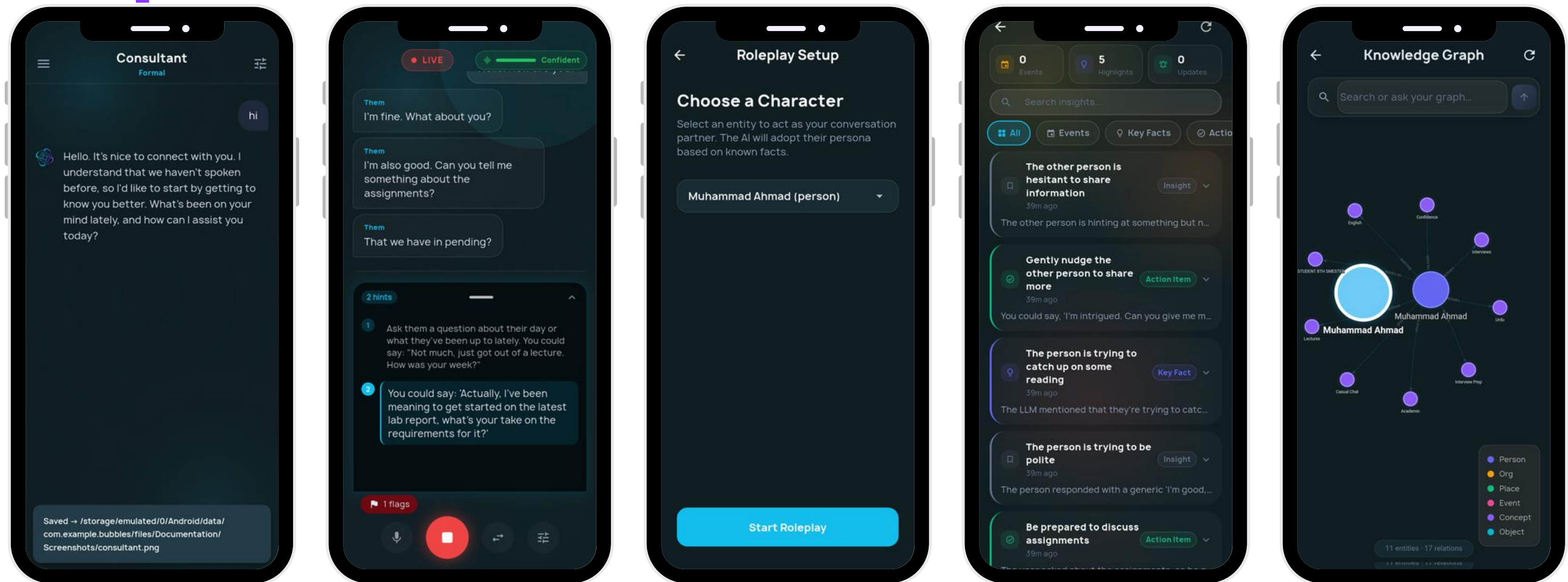


## Glassmorphic Visual Identity

Employs frosted glass effects and layered UI components to create a modern, visually engaging mobile experience

# Implementation: Frontend

# Implementation: Frontend



# Implementation: Backend



## FastAPI

Asynchronous high-performance framework handling concurrent WebSocket audio streams



## Multi-LLM Routing

Dynamic fallback logic between Cerebras (Low Latency) and Groq (Deep Reasoning)



## Task Queuing

14-worker ARQ queue on Redis for non-blocking extraction, sentiment, and summarization

# Custom websocket

## Low-Latency Transport

Ensures audio data is sent to the server with minimal delay

## High-Quality Capture sampled at 16,000Hz

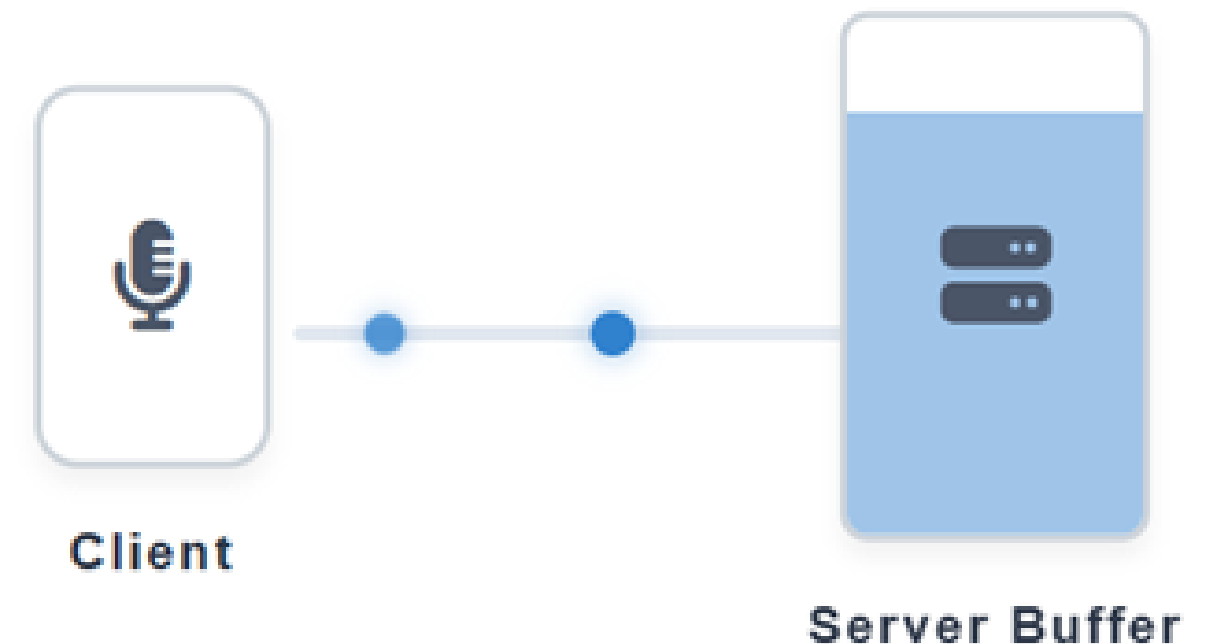
Captures clear, professional-grade audio for accurate transcription

## Adaptive Streaming

data transfer rates dynamically to maintain a stable connection

## Scheduled Processing with 12 second flush

Processes audio in reliable chunks to maintain real-time conversational flow



# Parallel Background Execution

## Pipeline Rearchitecture

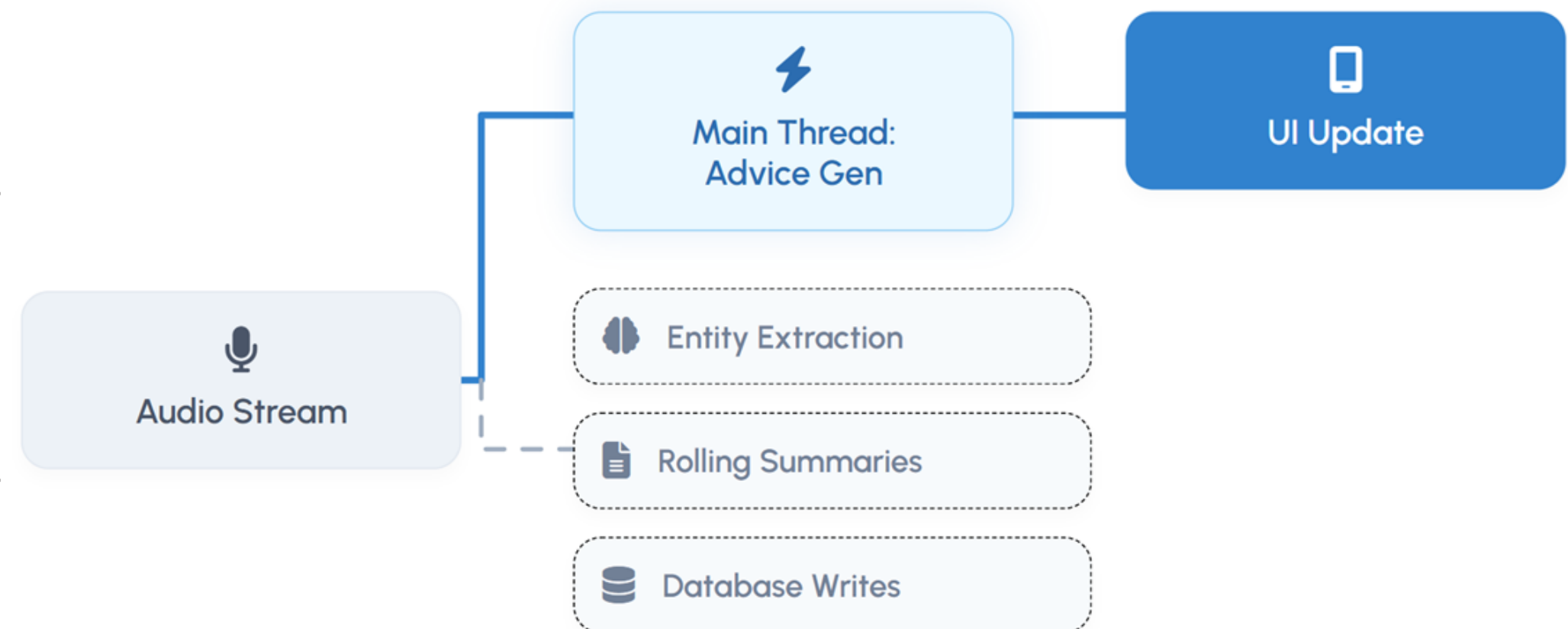
Splits operations into main and background threads for maximum efficiency

## Latency-Critical Path

Keeps urgent tasks like audio and UI on main thread for fast feedback

## Async Worker Offloading

Moves heavy processing and database writes to the background to prevent lag



# Strict Voiceprint Filtering

## Diarization Architecture: SpeechBrain ECAPA-TDNN

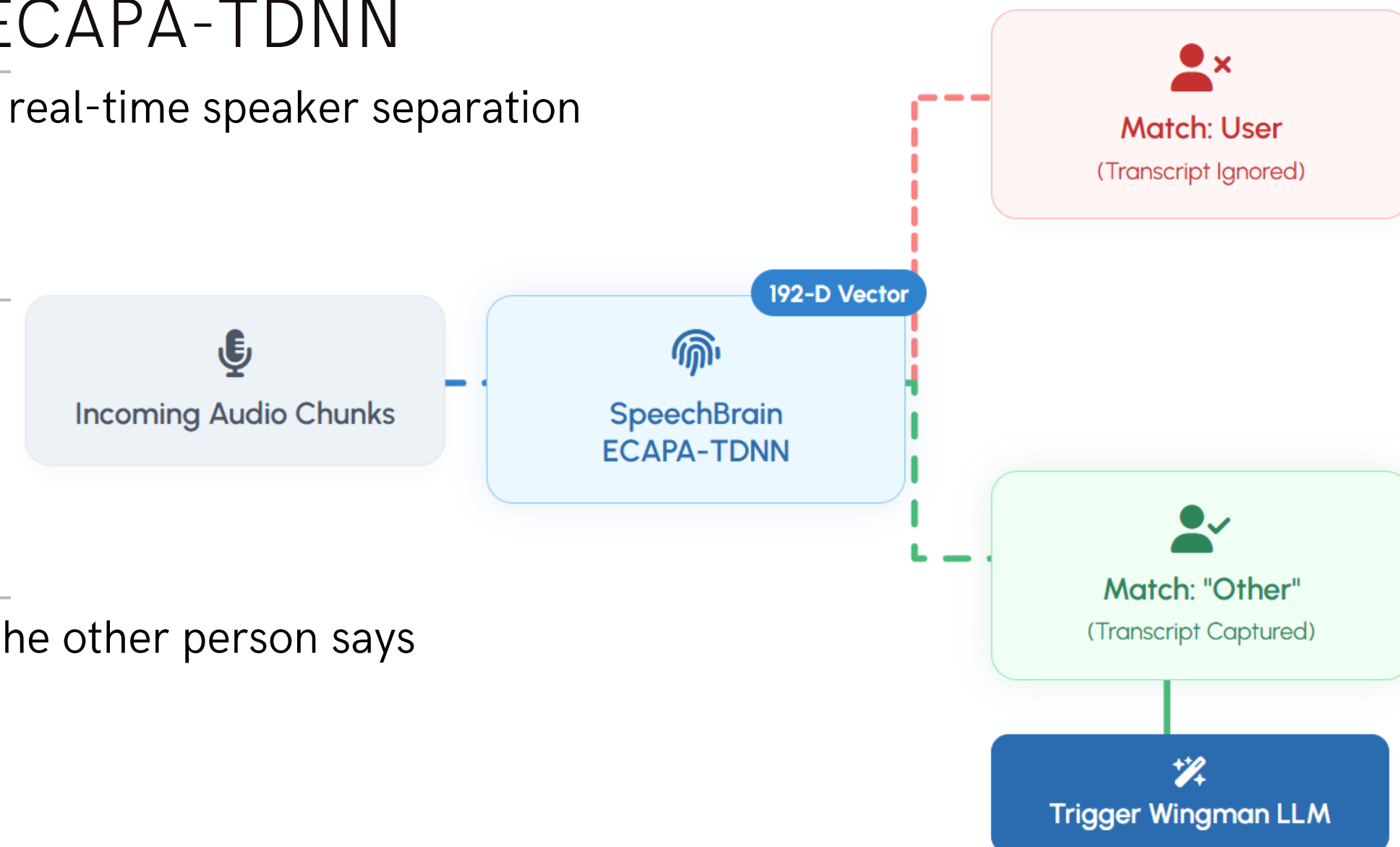
Integrates SpeechBrain ECAPA-TDNN for highly accurate, real-time speaker separation

## Vector Verification

Uses cosine difference to confirm the speaker's identity confidently

## Targeted Execution on other speaking

Ensures Wingman advice is triggered specifically by what the other person says



# Tiered Session Management

## Dual-Tiered Architecture

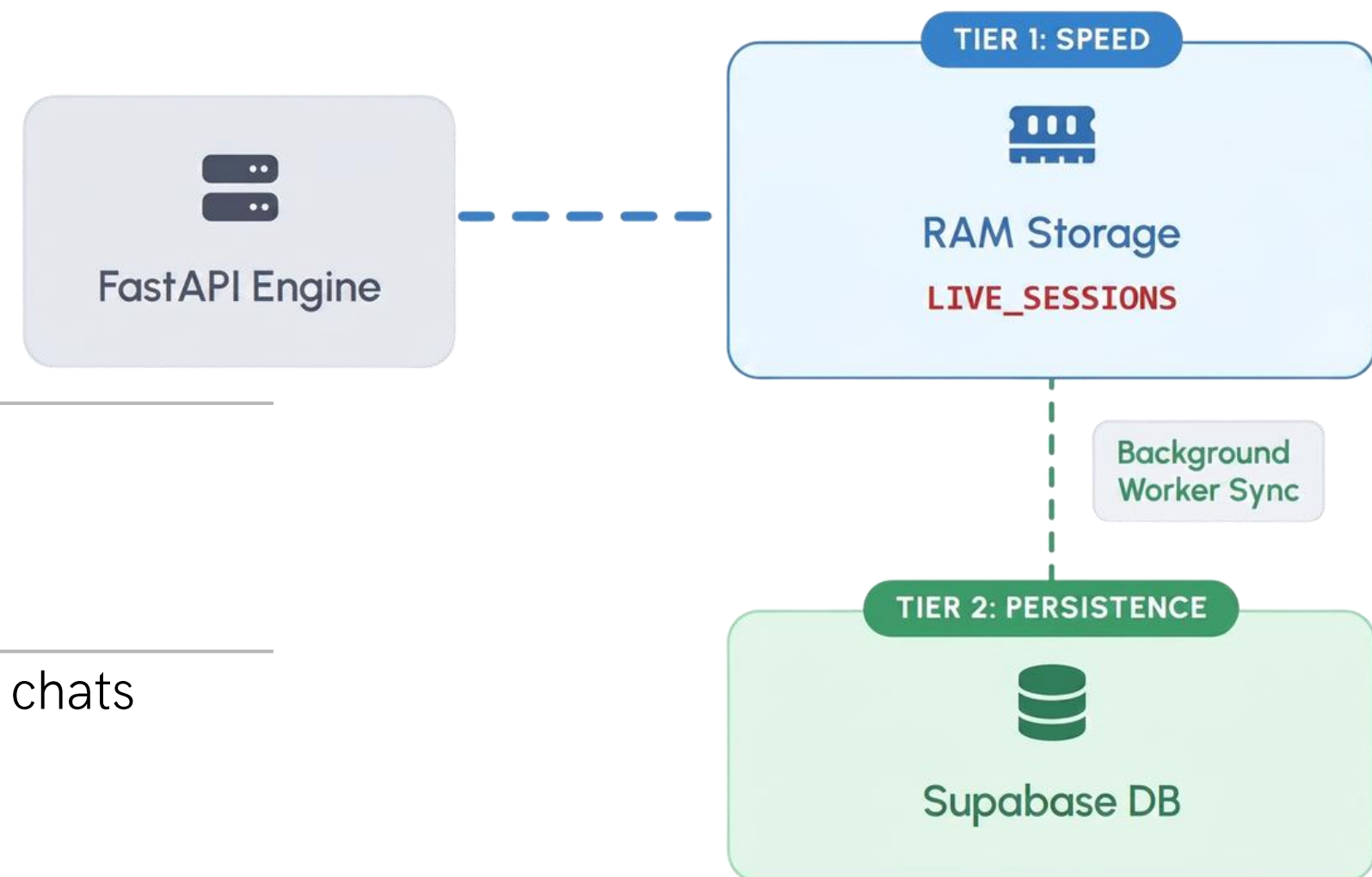
Splits data between a fast cache and permanent storage

## In-Memory Speed

Uses high-speed RAM for instant data access during active chats

## Reliable Persistence

Securely auto-saves completed sessions to prevent any data loss



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# 6.0 Results & Analysis

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# Testing setup

## Backend: Laptop with Nvidia 3060 mobile

Runs locally on a laptop powered by an Nvidia 3060 mobile GPU for fast, on-device AI processing

## Stack: FastAPI, Docker, Redis, Supabase

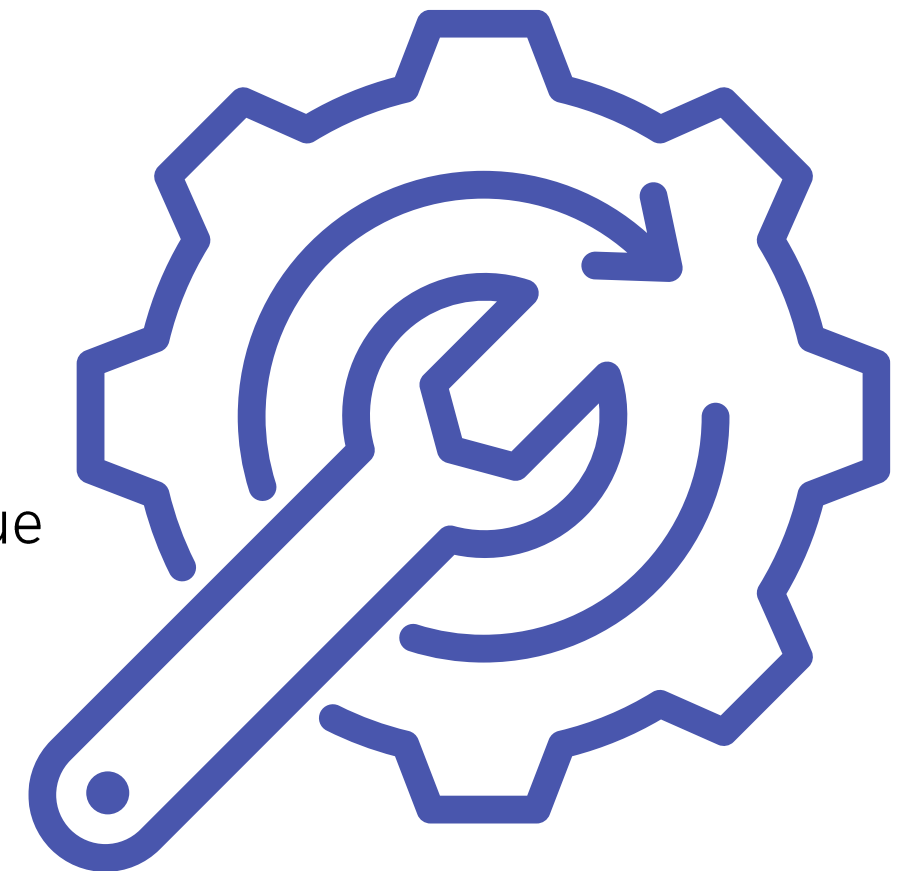
Utilizes scalable, backend architecture featuring FastAPI, Docker, Redis, and Supabase

## Dataset: 20+ Sessions, 3500+ Lines

Evaluated against test set of over 20 live conversational sessions and 3,500+ lines of dialogue

## Scale: 700+ Knowledge Graph Entities

Successfully manages and queries complex relational data within a Knowledge Graph of over 700 distinct entities



# ASR Accuracy (Groq Whisper)

Quiet Room :: 92%



Moderate Noise :: 85%



3+ Speakers :: 88%



Outdoor (Wind/Traffic) :: 80%



# Identity Aware Diarization

94%

Speaker  
Identification

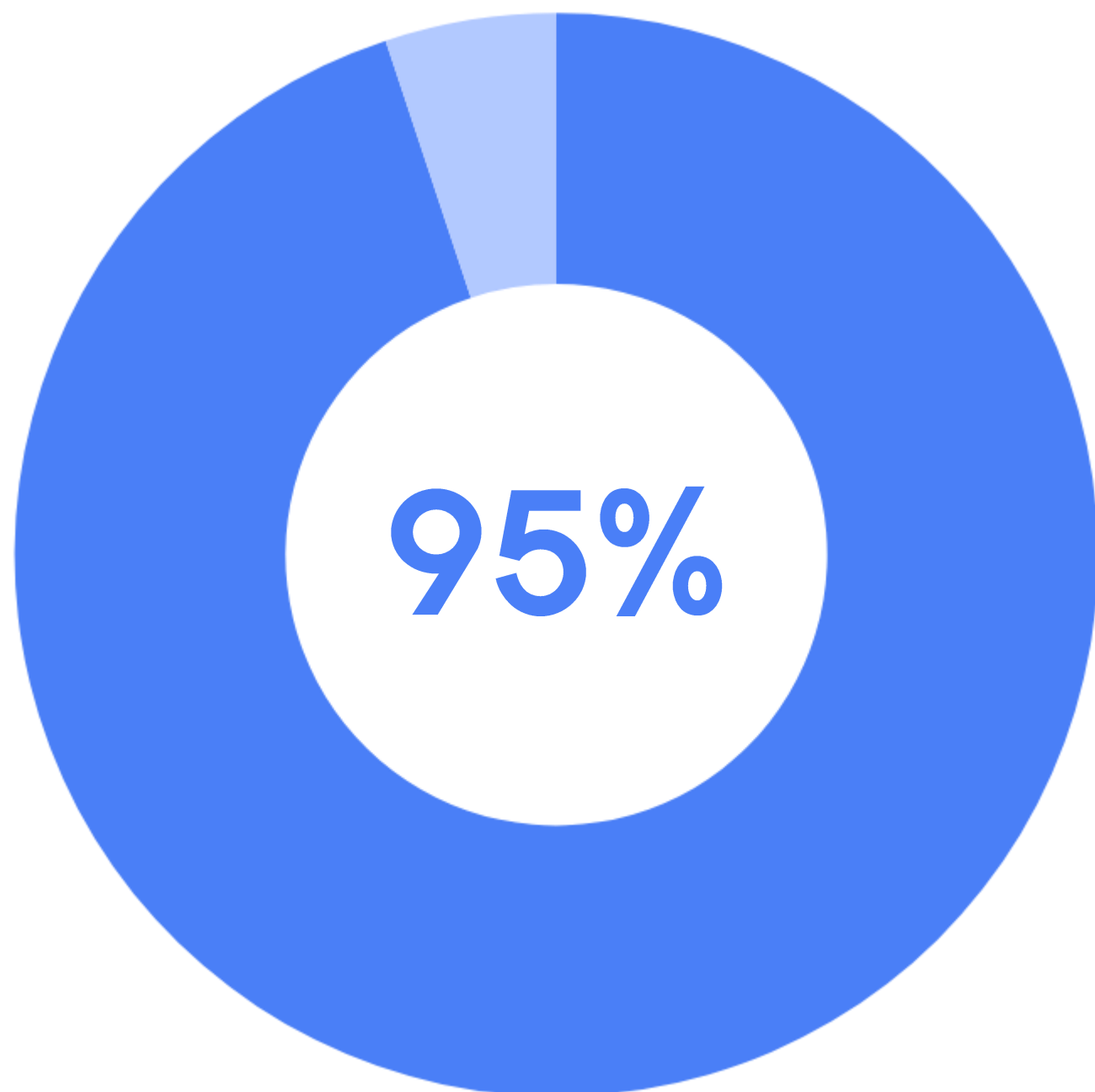
50ms

Voiceprint  
Computation

3%

False Positive Rate

# Hybrid RAG Retrieval Performance



**Hybrid (Vector Search + KG): 95%**

Vector Search Only: 72%

Graph Only: 68%

Scale: 700+ Knowledge Graph Entities

# Summary of System Execution



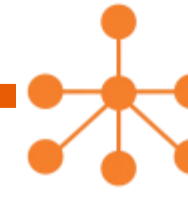
## Multi Modal

5 SOTA models  
orchestrated for  
specialized tasks:  
Llama 3.1 3b/70b, Groq  
Whisper, all-MiniLM and  
Picovoice.



## Parallelized

Context building moved to  
async background threads  
to zero-out latency.



## Context Aware

384-dim embeddings  
paired with relational  
graph neighborhood  
traversal.

# Competitive Analysis

| Feature              | Otter.ai              | Yoodli           | Google Assistant | Bubbles (Ours)                 |
|----------------------|-----------------------|------------------|------------------|--------------------------------|
| Real-time coaching   | No                    | ⚠ Partial        | No               | ✓ Yes (Wingman)                |
| Identity Diarization | Speaker 0/1           | Single speaker   | N/A              | ✓ Voiceprint                   |
| Cross-session memory | Per-meeting           | Metrics only     | ⚠ Partial        | ✓ KG + Vector                  |
| Gamification         | No                    | No               | No               | ✓ XP, Levels, Quests           |
| Privacy Model        | Enterprise harvesting | Cloud processing | Google ecosystem | ✓ Open-source, User-controlled |

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# 7.0 Future Work

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# Edge Computing

## Edge Processing

Shifting computational tasks directly to the mobile device instead of cloud

## Offline Functionality

Allowing app to provide coaching without an active internet connection

## Noise Filtering

Enhancing on-device audio cleanup before it reaches the AI models



# Multi-Speaker Diarization

## N-Speaker Tracking

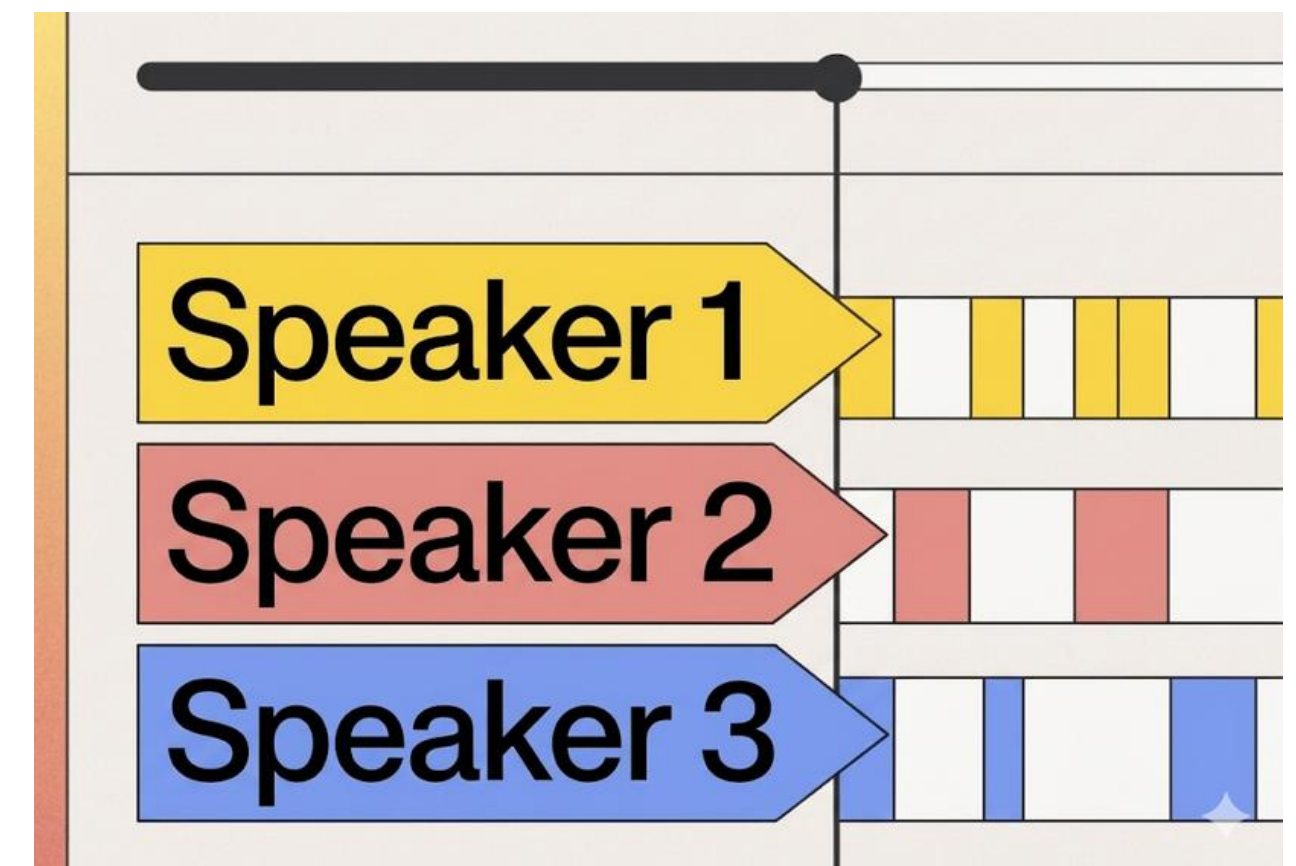
Expanding system to track and identify unlimited number of distinct speakers

## Manual Identity Tagging

Allowing users to manually assign names to newly recognized voices

## Interaction Maps

Visualizing who spoke to whom and how often during a meeting



# Multilingual Expansion

## Urdu STT Pipeline

Developing a dedicated speech-to-text pipeline for the Urdu language

## Contextual Retrieval

Adapting the RAG memory system to accurately fetch context on urdu language

## Broader Reach

Expanding the app's accessibility to non-English speaking demographics



# Multimodal Integration

## Video Feed Analysis

Processing camera input to analyze visual cues alongside audio

## Facial Expression Recognition

Gauging the other person's emotions through their facial reactions

## Speaking pitch and tone

Analyzing vocal inflections for deeper, more accurate sentiment analysis

## Holistic Metrics

Combining audio, visual, and textual data for a comprehensive communication score



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# 8.0 Conclusion

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# Conclusion

## Low Latency for advice

Successfully achieved sub-second response times for real-time coaching

## Accurate transcription

Maintained high transcription accuracy even in moderate noise environments

## Context-aware retrieval

Achieved ~95% relevance using the Hybrid Graph-RAG system

## Identity-Aware Diarization

Reliably distinguished the user's voice from others using precise vector matching



# Thank You

Group Members

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Supervisor:

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